Sequences, Series and the Binomial Theorem — Diagnosing Index, Common-Ratio and Convergence Errors

Algebra 2

Three errors account for most wrong answers in this unit. Each problem below contains one of them.

- **Index off by one:** $a_n = a_1 + (n-1)d$ and $a_n = a_1 r^{n-1}$ both step $n-1$ times, because the first term is already there before any step is taken.
- **Ratio read as a difference:** r is found by *dividing* consecutive terms, and one pair is never enough — check a second pair.
- **Convergence ignored:** $S_\infty = \dfrac{a_1}{1-r}$ is valid only when $|r| < 1$. Used outside that range it returns a confident number for a sum that does not exist.
- In each problem: name the error in words, then show corrected work.

1. Find and fix. An arithmetic sequence has first term $a = 7$ and common difference $d = 4$. Dev wants the 12th term and writes $a_{12} = 7 + 12(4) = 55$.

Say in one sentence why Dev counts one step too many, then compute the correct 12th term.

Answer: __________________

2. Find and fix. A geometric sequence begins 5, 10, 20, … Kai finds the common ratio by subtracting consecutive terms and reports $r = 5$.

(a) Find r correctly from the first pair of terms.
(b) Confirm it on the second pair. Say why checking one pair is not enough to identify a sequence as geometric.

Answer: ________________

3. Find and fix. Consider the arithmetic series whose first term is 3 and whose common difference is 5, summed over its first 10 terms.

Ines writes $S_{10} = \frac{10}{2}\big(3 + [\,3 + 5(10)\,]\big) = 280$.

Her formula is right and her last term is wrong. Name the slip, then compute the correct sum.

Answer: ________________

4. Binomial expansion. Expand $(x+2)^4$ completely, writing the terms in descending powers of x.

Then check your work with the index rule: in the expansion of $(x+p)^n$, the exponents on x and on p must add to n in every single term.

Answer: ________________

5. Find and fix. An infinite geometric series has first term 6 and common ratio $r = \frac{3}{4}$. Nia writes $S = 6 \div \frac{3}{4} = 8$.

(a) State whether this series converges, and why.

(b) Name the part of the formula Nia mis-copied, then compute the correct sum.

Answer: ________________

6. Find and fix. An arithmetic sequence starts at 7 and increases by 4 each step. A student wants to know which term equals 91, sets up $7 + 4n = 91$, and reports the answer. The equation is missing one adjustment. Write the correct equation, solve it, and state which term of the sequence equals 91.

Answer: ________________

7. Find and fix. In the expansion of $(x + 3)^5$, Vic wants the coefficient of the x^2 term. He reasons: "the power on x is 2, so $k = 2$," and computes $\binom{5}{2}3^2 = 90$.

(a) Use the rule that the exponents on x and on 3 must add to 5 to find the correct value of k.

(b) Compute the correct coefficient of the x^2 term.

Answer: ________________

8. Explain.

(a) Explain why an infinite geometric series has a finite sum only when $|r| < 1$. Refer to what happens to r^n as n grows.

(b) A classmate applies $S = \dfrac{a_1}{1-r}$ to a series of positive terms whose ratio is 2 and reports a *negative* sum. Explain, without computing anything, why that answer is impossible and what it reveals about the formula's condition.

(c) Write two checking habits: one that catches an off-by-one index, one that catches a ratio read as a difference.